

# PACKPROOF

## 1. Purpose of This Document

This document explains the complete power-control and shutdown-safe PCB design used in the PackProof system.

## 2. System Overview (Hardware)

The PackProof power system consists of:

5V Adapter

UPS Module + Li-po Battery

Relay-based Power Isolation Circuit

Raspberry Pi

GPIO-based Power Loss Detection

Key Objectives Achieved

Auto-power ON when adapter is connected

Safe shutdown on power loss

Zero battery drain after shutdown

Auto-restart when power returns

No accidental reboot during operation

## 3. Major Components Used

| Component  | Description                |
|------------|----------------------------|
| UPS module | 5V UPS with IN+,OUT+,B+,B- |
| Battery    | 1100mAh Li-Po              |
| Relay      | 5V SPDT relay              |
| Transistor | BC547 (NPN)                |

|            |                                                               |
|------------|---------------------------------------------------------------|
| Diode      | 1N4007 (flyback), 1N4148 / 1N4007 (GPIO isolation)            |
| Resistors  | 4.7k $\Omega$ , 2.2k $\Omega$ , 100k $\Omega$ , 7.3k $\Omega$ |
| Controller | Raspberry Pi Zero 2W                                          |

#### 4. UPS Module Connections

Adapter Input

Adapter +5V → UPS IN+

Adapter GND → UPS IN-

Battery Input

Battery + → UPS B+

Battery – → UPS B-

Output

UPS OUT+ → Relay (NO contact)

UPS OUT– → Raspberry Pi GND

#### 5. Relay-Based Power Control Circuit

##### 5.1 Relay Coil Connections

Relay Coil + to UPS OUT+

Relay Coil - to NPN Transistor Collector

##### 5.2 Relay Contact Configuration

Relay Common (C) to Raspberry Pi +5V (GPIO power rail)

Relay Normally Open (NO) to UPS OUT+

Relay Normally Closed (NC) – Not used

When GPIO pulls relay LOW: Relay energizes, connects adapter power to Pi

When GPIO pulls relay HIGH: Relay de-energizes, disconnects power

##### 5.3 Transistor & Control Circuit

GPIO27 to Base (through 10k resistor)

Collector to Relay Coil -

Emitter to GND

Diode across coil for flyback protection

## **6. GPIO Power Loss Detection**

GPIO26 monitors mains power via voltage divider:

Adapter +5V --- 10k resistor --- GPIO26 --- 10k resistor --- GND

When powered: GPIO26 reads 3.3V (HIGH)

When unplugged: GPIO26 reads 0V (LOW)

Interrupt triggers on falling edge to power\_monitor.py detects loss

UPS battery maintains 5V during safe shutdown sequence

## **7. Power Failure Handling Sequence**

Step 1: Power loss detected on GPIO26 (HIGH to LOW transition)

Step 2: Immediate UPS switchover (automatic, <100ms)

Step 3: Warning dialog displays on touchscreen

Step 4: Video recording is safely stopped

Step 5: File system is synced (fsync)

Step 6: GPIO27 pulled LOW to power-down relay

Step 7: Clean shutdown completes within 5-10 seconds

Step 8: Power isolated - battery drain prevented

UPS maintains system voltage throughout entire sequence

## **8. Automatic Power-ON Circuit (After Power Return)**

When adapter power is re-connected:

- Mains voltage appears at UPS IN+
- UPS detects mains and begins charging battery
- Relay remains in OFF state until manually activated
- GPIO27 must be set LOW to engage relay
- Power applied to Raspberry Pi GPIO pins
- System boots automatically with systemd launcher

## 9. Resistor Divider Calculation (GPIO26)

Provides 3.3V logic signal from 5V mains detect:

$$V_{out} = V_{in} \times (R_2 / (R_1 + R_2))$$

$$V_{out} = 5V \times (10k / 20k) = 2.5V$$

Actual value depends on component tolerances. Adjust R1 or R2 for precise 3.3V.

Use 1% tolerance resistors for accuracy.

### 9.1 Resistor Divider Naming

R1 = Upper resistor (toward +5V)

R2 = Lower resistor (toward GND)

Both values typically 10k ohm

### 9.2 GPIO Logic Interpretation

HIGH (3.3V) = Mains powered

LOW (0V) = Battery backup active

## 10. Power Loss Behavior

UPS Switchover Time

<100 milliseconds from power loss to battery supply

System has ample time for:

- Detecting GPIO26 low state
- Triggering warning dialog
- Stopping video recording
- Syncing file system
- Initiating clean shutdown

Battery provides smooth transition - no brownout or data corruption